

## RESEARCH ARTICLE



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## Temporal gene expression profiling of maslinic acid-treated Raji cells†

Wai Meng Lau,<sup>a</sup> Menaga Subramaniam,<sup>id a</sup> Hoe Han Goh<sup>id b</sup> and  
Yang Mooi Lim<sup>id \*ac</sup>

Maslinic acid is a novel phytochemical reported to target multiple signaling pathways. A complete gene expression profile was therefore constructed to illustrate the anti-tumourigenesis effects of maslinic acid in Raji cells across five time-points. Microarray analysis was used to identify genes that were differentially expressed in maslinic acid treated Raji cells at 0, 4, 8, 12, 24 and 48 h. Extracted RNA was hybridized using the AffymetrixGeneChip to obtain expression profiles. A total of 109 genes were found to be significantly expressed over a period of 48 hours. By 12 hours, maslinic acid regulates the majority of genes involved in the cell cycle, p53 and NF- $\kappa$ B signaling pathways. At the same time, XAF1, APAF1, SESN3, and TP53BP2 were evidently up-regulated, while oncogenes, FAIM, CD27, and RRM2B, were down-regulated by at least 2-fold. In conclusion, maslinic acid shows an hourly progression of gene expression in Raji cells.

## Introduction

Maslinic acid, a pentacyclic triterpene (Fig. 1), has garnered intense research attention lately due to its *in vivo* and *in vitro* effects on tumour prevention<sup>1</sup> in colon cancer cells,<sup>2</sup> melanoma cells<sup>3</sup> and astrocytes.<sup>4</sup> Proteomic studies revealed the effect of maslinic acid in activating the cellular apoptotic mechanism in HT-29 and Raji cancer cells in a time and concentration-dependent manner.<sup>5,6</sup> Besides, maslinic acid was also found to be able to stabilize the degradation of Nrf2 protein in the HepG2 cell line.<sup>7</sup> Nrf2 is the chief transcription factor involved in defusing cellular stress caused by reactive oxidative species and its activation would subsequently lead to the activation of the Nrf2-Keap1-ARE pathway; one which plays a pivotal role in the cellular defense mechanism.<sup>8</sup> Apart from this pathway, maslinic acid was also reported to inhibit the NF- $\kappa$ B signaling pathway in a time-dependent manner. The NF- $\kappa$ B pathway regulates the expression of numerous genes and it is particularly

important to B-cell biology, as well as cancer development. The overexpression of genes involved in the NF- $\kappa$ B pathways often contributes to neoplasia; in this study maslinic acid has prevented this event through the suppression of p65 binding activity.<sup>6</sup> Evidently, this compound was shown to be cell-type specific and acts against multiple pathways.<sup>9</sup> It was postulated that maslinic acid targets the G0/G1 phase of the cell cycle and augments caspase-3 activities independent of the p53 mechanism.<sup>10,11</sup> Other noteworthy attribute of maslinic acid includes low toxicity profile in animal studies,<sup>9,12</sup> anti-inflammation<sup>13</sup> and inhibition of cellular proliferation capabilities.<sup>14</sup> Collectively, these findings suggest the potential of transforming maslinic acid into a tumour suppressant drug.

The effects of maslinic acid against cancer were evidently documented at the proteomic level in HT-29 and Raji cancer

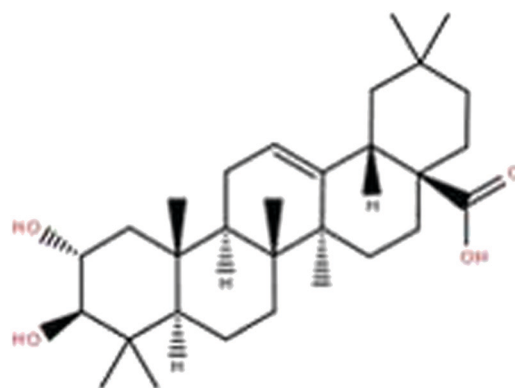


Fig. 1 Pentacyclic structure of maslinic acid.

<sup>a</sup> Centre for Cancer Research, Faculty of Medicine and Health Sciences, Universiti Tunku Abdul Rahman, PT21144, Jalan Sungai Long, Bandar Sungai Long, 43000 Kajang, Selangor, Malaysia. E-mail: ymlim@utar.edu.my; Fax: +603 90191959; Tel: +603 90194722 etxn 178

<sup>b</sup> Institute of Systems Biology, Universiti Kebangsaan Malaysia, UKM Bangi, Bangi, Malaysia

<sup>c</sup> Department of Pre-Clinical Science, Faculty of Medicine and Health Sciences, Universiti Tunku Abdul Rahman, Lot PT21144, Jalan Sungai Long, Bandar Sungai Long, 43000 Kajang, Selangor, Malaysia

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cells, but there is still an information gap on the global gene expression profile in these cells.<sup>2,5,6</sup> The information of the global gene expression is highly essential for a better understanding of its molecular mechanism and to elucidate its anti-tumour promoting mechanism in these cells. Hence, a temporal gene expression was constructed in this study aiming to provide a birds-eye view of the signaling networks regulated by maslinic acid in preventing tumour promotion in Raji cells. This gene expression profile would serve as a reference point for future studies where subsequent investigations upon maslinic acid can be focused onto the pathways in which selected genes are significantly regulated by maslinic acid in Raji cells.

## Materials and methods

### Cell culture

Raji cells, derived from the human lymphoblastoid cell line, were obtained from RIKEN cell bank. The cells were maintained in commercial RPMI 1640 medium supplemented with 10% fetal bovine serum (GIBCO, South America) at 37 °C in a humidified incubator circulated with 5% carbon dioxide. Only cultures with more than 90% cell viability as determined by a trypan blue exclusion method were used for the anti-tumour promoting assay.

### Chemicals

1 mg of maslinic acid ( $\geq 98\%$  purity) (Santa Cruz Biotechnology Inc., CA) was dissolved in 1 mL of dimethyl sulfoxide (DMSO) (Fisher Scientific, MA). Phorbol 12-myristate 13-acetate (PMA), sodium *n*-butyrate (*SnB*) and fluorescein isothiocyanate were purchased from Sigma Chemical Co. (MO).

### Anti-tumour promoting assay

$1 \times 10^6$  cells per mL of Raji cells were placed in a 96-well plate incubated with 3 mM of sodium *n*-butyrate, 0.05  $\mu$ M phorbol 12-myristate 13-acetate (PMA) and 19  $\mu$ M maslinic acid in RPMI 1640 media supplemented with EBV-infected serum. The cell cultures were prepared in triplicate and the cells were maintained at 37 °C in a 5% CO<sub>2</sub> atmosphere. The total RNA and proteins were harvested at six time-points at 0, 4, 8, 12, 24 and 48 hours. The samples were processed immediately with a Total RNA/Protein extraction kit (Macherey-Nagel, Germany) as per the manual's instruction. The purity and integrity of total RNA extracted were determined by using a Nanodrop<sup>®</sup> UV-Vis Spectrophotometer (Implen, Germany) and 1.2% denaturing agarose gel respectively. Only intact RNA samples with a purity of  $A_{260/280}$  1.9–2.0 were used for downstream experiments. Eluted RNA and solubilised proteins were stored in aliquots at –80 °C until further use.

### Probe hybridization

20 ng  $\mu$ L<sup>–1</sup> of good quality RNA was used to generate biotin-labeled cDNA with the Applause WT-Amp ST System (NuGEN, CA) kit and the samples were subsequently purified using a MinElute Reaction Cleanup kit (Qiagen, Germany) according to the manufacturer's instructions. Following purification, 22.7 ng  $\mu$ L<sup>–1</sup> of

cDNA was hybridized for 14 hours on the AffymetrixGeneChip<sup>™</sup> (Affymetrix, CA) and the arrays were scanned with the GeneChip Scanner 3000 7G (Affymetrix, CA). The experiment was performed in triplicate.

### Statistical methods

All raw data were processed with GeneSpring GX12 (Agilent Technologies, CA) *via* the exon expression default analysis settings. The data were grouped according to its respective condition in triplicate and subjected to a robust multichip average 16 (RMA16) pre-processor to obtain a RMA-normalized signal value for all arrays. The dataset was tested with different permutations of *p*-value cut-off and multiple testing corrections as an attempt to select the best combination analysis method. One-way ANOVA was selected as the preferred method to examine the differences between mean expression levels and these values were corrected with the Bonferroni correction method. A *p*-value of less than 0.05 was used to discriminate between differentially expressed genes. Selected genes which were significantly expressed across two time-points were analyzed with the DAVID bioinformatics tool for pathway elucidation purposes. The microarray data were made accessible through GEO Series accession number **GSE39149** (<http://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE39149>).

### Validation by real-time qPCR

A probe-based real-time PCR format was used for this purpose. Each reaction contains 1  $\mu$ L of cDNA aliquot and 9  $\mu$ L of Solaris qPCR mastermix (Thermo Scientific, USA) with 800 nM of forward and reverse primers each and 200 nM of gene specific probe (List of probes: GCCAAGGTCATCCATGA (GAPDH), TGGACGGAAAACAGTCTTA (OAS2), TCCATGTCGGTGTCAGAGC (ISG15), TCCATGTCGGTGTCAGAGC (XAF1), TCAGCACCAAC CAAGGG (CXCL9), TGAGGAACTGGTGCAAT (IF144L), AAAA TACTTTGAAAGAC (DDX58). The real-time PCR assay was performed using a RotorGene thermal-cycler (QIAGEN, Germany). The cycling conditions comprise a 15 minute enzyme activation step at 95 °C, followed by 40 cycles of denaturation step at 95 °C for 15 seconds and an annealing/extension step at 60 °C for 60 seconds. The expression of the selected genes was normalized to a housekeeping gene, GAPDH. The amplification was performed in triplicate and the procedure conforms to the MIQE checklist.

### Western blot analysis

STAT-1 and TNF- $\alpha$  proteins were selected based on a microarray study. Protein expression was validated using western blot. The extracted protein was quantified using a Bio-Rad DC protein assay kit (Bio-rad, CA). 50  $\mu$ g of total protein was loaded onto the SDS-PAGE (12% resolving and stacking gel) and the electrophoresis procedure was run at 110 V for 60 minutes. The separated proteins were transferred onto a PVDF membrane (Pierce, MA) at 15 V for 16 minutes. The membrane was blocked with Starting Block (TBS) blocking buffer (Pierce, MA) for 30 minutes and subsequently hybridized overnight at 4 °C with an appropriate amount of primary antibodies. The blots were then washed for 5 minutes thrice with TBS-T prior to one-hour incubation with HRP-conjugated secondary antibody and

followed by chemiluminescent detection (Cell Signaling Technology, MA) with UVP gel imager. Equal loading of proteins was confirmed by probing the same protein samples with loading control, anti-GAPDH. Monoclonal mouse antibodies were used, and the antibodies were anti-STAT1, anti-TNF- $\alpha$  and horseradish peroxidase (HRP) conjugated mouse anti-IgG.

Results

This study employed Raji cells as a cell model to investigate the inhibitory ability of maslinic acid against EBV early antigen induction. In this study, a temporal gene expression profile was created and aimed to capture events occurring as early as 4 hours after PMA and *S*nB stimulation until the peak hour of EBV-EA induction by 48 hours (against 0 h as baseline). In the preparation for microarray data analysis, the GeneChip® Operating Software (Affymetrix Inc., CA) was used to facilitate image acquisition and to perform initial data analysis which includes data quality assessment.

Approximately 28 870 known human genomes were analyzed using the AffymetrixGeneChip Human Gene 1.0 ST array (Affymetrix Inc., CA) and gene transcripts with a *p*-value of less than 0.05 and expression fold-change of  $\pm 2.0$  across at least 2 time-points were considered as significant (Table S1: ESI†). These criteria yielded a total of 215 significantly regulated genes which were sorted into 12 broad pathways by DAVID Bioinformatics tools (Table 1) namely, B cell receptor signaling, cell cycle and cellular proliferation, DNA repair, NF- $\kappa$ B, MAPK signaling, TRAF signaling, RAS signaling, RIG-1-like receptor signaling, TGF- $\beta$ , MHC antigen processing and presentation, and p53 signaling pathways.

This study provides a neutral observation of all gene transcripts and the sequential expression of the genes throughout the experimental period from 0 to 48 hours have shown a distinct gene regulation pattern in terms of the number of genes being regulated before and after 12 hours of maslinic acid treatment (Fig. 2). Hence, a comparison was made between time-points at 4 hours (early EA-induction phase), 12 hours (mid EA-induction phase) and 48 hours (late EA-induction phase) to identify the persistently regulated genes (Fig. 3). 109 genes were discovered

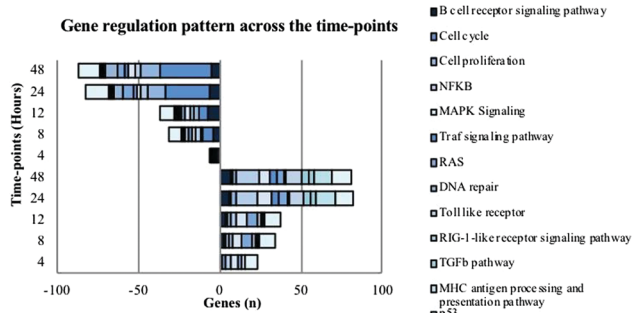


Fig. 2 The gene regulation pattern of significantly expressed genes ( $p \leq 0.05$  and fold-change cut-off at 2.0) across time-points. Negative and positive values denote down-regulation and up-regulation of gene expression respectively.

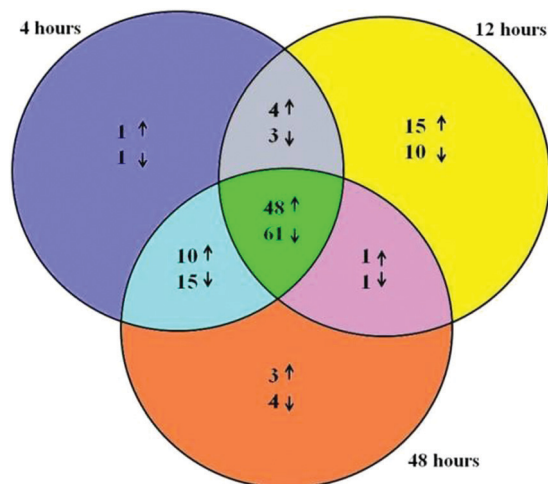
to be commonly regulated across the time-points in response to the treatment with maslinic acid at 19  $\mu$ M (Table S2: ESI†)

It was apparent from the gene expression profile that maslinic acid inhibited the induction of EBV-EA *via* several pathways and emphasis was placed onto the cell cycle, B-cell receptor signaling, Ras and p53 signaling pathways. Genes involved in these pathways were persistently regulated as time progresses. To better understand the functional relationships between these genes, KEGG pathway analysis was performed which showed that the altered genes were part of the core network of genes driving B-cell receptor (Fig. S1: ESI†), cell cycle (Fig. S2: ESI†), DNA replication (Fig. S3: ESI†) and MHC1 antigen presentation and processing pathway (Fig. S4: ESI†).

Six genes that were significantly expressed in the pathway of interest were selected for result validation purposes. Fold-change was derived from the mean expression value of the time-points (4, 8, 12, 24 and 48 hours) against the control group (0 hour). Log-transformed expression ratios were obtained from the six genes and these values were plotted against its corresponding values from the microarray data. The ability of maslinic acid in stimulating cell death was evident from the prominent up-regulation of STAT-1 and XAF1; both of which contribute greatly towards cell death (Fig. 4). The measurements of gene expression determined by the microarray system and qPCR method showed

Table 1 Filtered gene list analyzed using a DAVID Bioinformatics tool to identify pathways regulated by the treatment with maslinic acid

| Time-point (h)                                  | Expression (number of genes) |   |    |    |    |                 |   |    |    |    |
|-------------------------------------------------|------------------------------|---|----|----|----|-----------------|---|----|----|----|
|                                                 | Up-regulation                |   |    |    |    | Down-regulation |   |    |    |    |
|                                                 | 4                            | 8 | 12 | 24 | 48 | 4               | 8 | 12 | 24 | 48 |
| B cell receptor signaling pathway               | 2                            | 3 | 4  | 6  | 7  | 2               | 4 | 7  | 6  | 5  |
| Cell cycle and cellular proliferation           | 2                            | 3 | 3  | 4  | 3  | 1               | 8 | 9  | 29 | 43 |
| DNA repair                                      | 0                            | 0 | 0  | 1  | 1  | 0               | 0 | 1  | 6  | 7  |
| NF- $\kappa$ B pathway                          | 3                            | 2 | 3  | 13 | 14 | 0               | 3 | 2  | 4  | 4  |
| MAPK signaling pathway                          | 4                            | 6 | 7  | 9  | 7  | 0               | 2 | 3  | 3  | 4  |
| TRAF signaling pathway                          | 0                            | 0 | 0  | 4  | 4  | 0               | 0 | 0  | 2  | 2  |
| RAS                                             | 3                            | 6 | 6  | 6  | 5  | 1               | 2 | 3  | 6  | 5  |
| Toll-like receptor pathway                      | 2                            | 2 | 2  | 9  | 10 | 1               | 2 | 1  | 1  | 1  |
| RIG-1-like receptor signaling pathway           | 0                            | 0 | 0  | 4  | 4  | 0               | 0 | 0  | 0  | 0  |
| TGF- $\beta$ pathway                            | 0                            | 1 | 2  | 3  | 3  | 0               | 1 | 1  | 1  | 1  |
| MHC antigen processing and presentation pathway | 0                            | 1 | 1  | 12 | 11 | 0               | 1 | 1  | 1  | 1  |
| p53 pathway                                     | 7                            | 9 | 9  | 11 | 12 | 1               | 8 | 8  | 14 | 14 |



**Fig. 3** Total genes regulated by the treatment with maslinic acid at three time-points representing the early, mid and late EA-induction phase (4 hours, 12 hours and 48 hours respectively). Down arrow and up arrow indicate down-regulation and up-regulation of gene expression respectively.

good correlation ( $R^2 \geq 0.70$ ) with the best correlation shown by XAF1 and CXCL9 ( $R^2 \geq 0.90$ ), followed by IF144L and OAS2 ( $R^2 \geq 0.80$ ), and DDX58 as well as ISG15 ( $R^2 \geq 0.70$ ) as shown in Fig. 5. TNF- $\alpha$  and STAT-1 proteins are the key players in alternative cell death mechanism and MHC1 antigen presentation and processing pathways, respectively. TNF- $\alpha$  and STAT-1 proteins are the key players in the alternative cell death mechanism and these proteins were selected based on their total gene expression level ( $\geq 3.5$ -fold,  $p$ -value  $< 0.01$ ) (Table S6: ESI $^\dagger$ ). The detection of proteins of interest was successful and this indicated that maslinic acid has effectively regulated cellular expression at both the genomic and proteomic stage (Fig. 6).

## Discussion

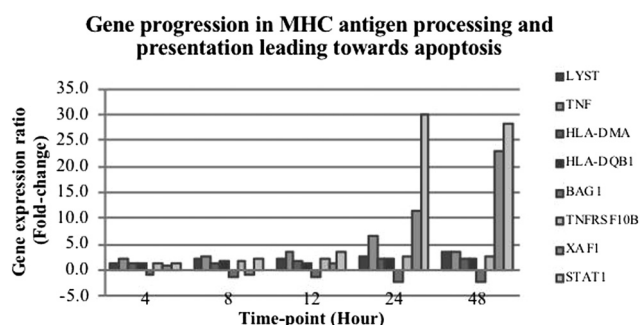
The mRNA expression profile has shown that the early EA-induction phase involved genes which were mainly affected by the application of PMA and SnB. This phase of EA-induction was reflected through the up-regulation of *RGS2*, *NFAT5* and *VIM*; all of which were stimulated by PMA. *RGS2* and *NFAT5*

were up-regulated in the first 4 hours, followed by *VIM* in the next 4 hours and constitutive down-regulation of *BCL7A*, *CR2*, *BLNK*, *PLEK*, *ITGB3BP*, and *CD22* during the same incubation period of 8 hours (Table S3: ESI $^\dagger$ ). To better understand the functional relationships between these genes, we performed KEGG pathway analysis (Fig. S1: ESI $^\dagger$ ), which showed that the altered genes were part of the core network of genes driving B cell receptor signaling.

It was proposed that these episodes of signal transmission have activated the MAPK signaling pathway resulting in significant up-regulation of several downstream factors; namely *FOS*, *RAPGEF2* and *RALB* within 8 hours (Table S6: ESI $^\dagger$ ). *RALB* is an oncogene that functions as an effector for the downstream Ras signaling pathway and this gene was persistently expressed which might involve the survival of the tumour cells. It was observed that *RASSF6* was reciprocally down-regulated and a crosstalk of signaling pathways was deduced from these events since *FOS* was also affected by the expression of *STAT3*.<sup>15</sup> *STAT3* is an important gene that provides a direct link between oncogenesis and immunosuppression. The up-regulation of *STAT3* favours not only tumour survival but also supports the regulation of TNF, survivin and c-MYC proteins through the NF- $\kappa$ B signaling pathway.<sup>16,17</sup> This pathway enhances cellular proliferation when it is activated alongside MAPK signaling pathways in a phospho-relay system.<sup>18,19</sup> This scenario presents EBV with an opportunity to infiltrate the immune system and subsequently augment the tumour promotion stage.

Quiescent cells are notoriously driven into proliferation by *MYC*, an oncogene that is over-expressed in many types of lymphoma.<sup>20,21</sup> However, this gene was down-regulated by approximately 2-folds and this expression level was sustained until the end of the experiment. For that reason, maslinic acid is suggested to have regulated the expression of *MYC* and concurrently be held responsible for the down-regulation of *IL7R* and *CR2* as well as the up-regulation of *APAF1*, *SESN3*, *TNFAIP3*, *PDP1* and *RYBP*. *CR2* was likely to be up-regulated in response to PMA<sup>22</sup> but this gene was down-regulated considerably by as much as 3-fold after 4 hours of treatment with maslinic acid (Table S3: ESI $^\dagger$ ). Taken together, the expression of these genes (*MYC*, *APAF1*, *RYBP*, *SESN3*, *IL7R* and *TNFAIP3*) depicted that 4 hours of treatment with maslinic acid would hamper the initial events of tumourigenesis by raising suppressing attempts *via* the up-regulation of both *SPRED2* and *TNFAIP3* and the down-regulation of several oncogenes (*BCL7A*, *FOS* and *MYC*). The tumour antagonistic effect; aimed at the Ras signaling pathway, was further fueled by the continuous down-regulation of *RASGRP1* and up-regulation of *ARHGEF12* after 8 hours of incubation with maslinic acid. *RASGRP1* is a gene which encodes for a Ras exchange factor and its expression is mediated in either a PKC-independent<sup>23</sup> or PKC-dependent<sup>24,25</sup> manner.

Furthermore, subsequent exposure to 19  $\mu$ M of maslinic acid for 8 to 12 hours has revealed that most genes targeted are involved in the p53 pathway. 18 genes in this pathway were differentially regulated, in which eight genes were down-regulated (*RNF144B*, *FAIM*, *DDB2*, *ZBTB2*, *AVEN*, *CD27*, *TP53*, *RRM2B*) and nine genes were up-regulated (*TP53BP2*, *APAF1*, *SERPINB9*, *SESN3*, *RYBP*, *PDP1*,



**Fig. 4** The progression of genes involved in the MHC antigen processing and presentation pathway across the time-points ( $p < 0.05$ , fold-change  $\geq 2.0$ ).



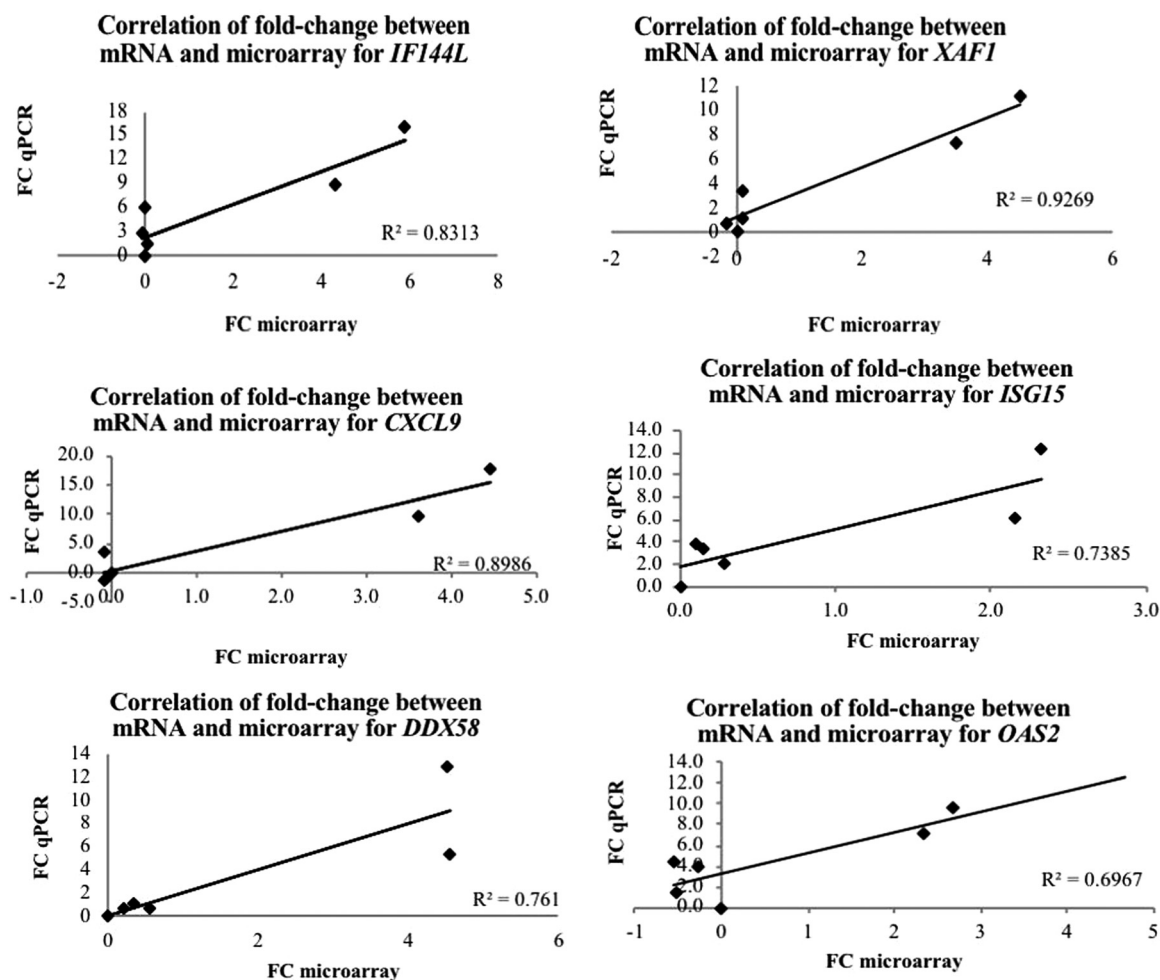


Fig. 5 Data obtained from the microarray experiment was validated with the qPCR method. The data show a correlation ( $R^2$  value) between 0.70 and 0.90 ( $p < 0.05$ ).

| Protein<br>(size)                 | Time-point (Hours) |             |             |
|-----------------------------------|--------------------|-------------|-------------|
|                                   | 4                  | 12          | 48          |
| (a) GAPDH<br>(37.0 kDa)           |                    |             |             |
| Expression value<br>(fold-change) | 1.0 ± 1.0          | 1.04 ± 0.05 | 1.04 ± 0.05 |
| (b) STAT1<br>(91.0 kDa)           |                    |             |             |
| Expression value<br>(fold-change) | 1.0 ± 1.0          | 0.9 ± 0.0   | 1.2 ± 0.0   |
| (c) TNF-α<br>(35.6 kDa)           |                    |             |             |
| Expression value<br>(fold-change) | 1.0 ± 1.0          | 0.9 ± 0.9   | 0.8 ± 1.9   |

Fig. 6 Maslinic acid regulated cellular expression at a protein level. TNF- $\alpha$  and STAT1 validate the activities of maslinic acid, which targets key regulatory proteins of multiple pathways in Raji cells.

*SERPINE1*, *LPIN1*, *SLC2A3*) (Table S6: ESI<sup>†</sup>). Notable activities in this pathway are the down-regulation of oncogenes; *FAIM* and *AVEN*, and the up-regulation of apoptosis inducing genes; *APAF1*

and *RYBP*. At the same time, it was observed that all phases of the cell cycle were gradually affected; beginning with significant suppression of cell cycle genes and followed by genes involved in DNA replication.

Cell proliferation is tightly associated with the DNA repair system and in EBV-infected B-cells, the virus would activate a series of DNA damage responses (DDRs) to prevent replicative stress and genomic instability; hence these DDRs would greatly facilitate efficient viral DNA replication.<sup>26</sup> This function is achieved with the expression of *RNASEH2A*, but the expression profile (Table S4: ESI<sup>†</sup>) suggested that the DNA repair system was deregulated given that genes associated with DNA repair, *DNA2*, *RPA2*, *RAD51* and *RAD51AP1*, were found to be down-regulated. DDRs and the cell cycle are regulated in parallel and as such, pre-replicative complexes in the G1 phase will be vigilantly scrutinized. The pre-replicative complexes are assembled from the origin recognition complex and MCM protein and its encoding genes are usually expressed at the G1/S boundary to propel the cell cycle into the S phase. Consequently, the down-regulation of *MCM4*, *MCM6*, *MCM10*, *ORC1*, *ORC6*, *CDC6*, *FOXN1* and *CDKN3* would cause the cell

cycle to be arrested at the G1/S phase. The G1/S phase is also affected by the down-regulation of *SKP2*, *CKS1*, *FOXM1* and *CDKN3*.<sup>27</sup> In addition, the down-regulation of *CDC25A* and *CDC45* by at least 2.5-folds also suggested that the cells were detained at S phase. Simultaneous down-regulation of genes which encode protein kinases, *WEE1*, *CCNA2*, *CCNE2*, *CENPE*, *CENPF*, *PLK1* and *TTK*, also prevented the cells from proceeding into the G2/M phase.

Another group of genes that responded to the treatment with maslinic acid are genes involved in the MHC antigen processing and presentation pathway as shown in Table S5: *ESI*† *PSMB9* and *TAP1* were down-regulated during the early and mid-EA induction phase but these genes were expressed by at least 2-folds after 24 hours. A defect in this mechanism would confer resistance towards apoptosis and ultimately augment tumour development by allowing tumour cells to evade immune surveillance.<sup>28,29</sup> In view of this, its expression is greatly appreciated because it can potentially outwit viral immune evasion approaches alongside the expression of *SQSTM1* and other cytokines by mediating alternative cell death pathways such as autophagy.<sup>30</sup>

This result is especially noteworthy because it has suggested a novel and yet to be discovered function of maslinic acid in re-instating the ability to present viral epitopes *via* the MHC antigen presenting mechanism in PMA-activated Raji cells. The up-regulation of these genes (*LYST*, *HLA-DMA*, *HLA-DQB1* and *TNF- $\alpha$* ) would have restored the cell's capability in presenting tumour-antigen derived peptide complexes onto the cell surface and subsequently facilitated immunosurveillance activities which eventually leads to apoptosis.<sup>31</sup> This mode of action could be triggered by *TNF- $\alpha$* , a cytokine found to have sensitized Raji cells to the apoptotic cell death mechanism.<sup>32,33</sup> The ability of maslinic acid in stimulating cell death was evident from the prominent up-regulation of *STAT1* and *XAF1*; both of which contribute greatly towards cell death (Fig. 4).

On the same note, it was observed that during the late EA-induction phase, several genes involved in the promotion of inflammation were remarkably up-regulated. *CCL3* is a strong inducer of *TNF- $\alpha$* <sup>34</sup> and this gene was expected to be specifically down-regulated by EBV, but it was highly expressed in unison with the up-regulation of *CXCL9*, *CXCL10* and *CXCL11* (*p*-value < 0.01, FC  $\geq$  6.7).

As EA-induction by PMA and *SnB* is progressively approaching its optimum state after a 24 hour incubation period, both *CASC5* and *BUB1* were significantly suppressed to hinder kinetochore formation and halted the cell cycle at the M phase. Proceeding cell cycle arrest, further disruptions aimed at cellular proliferation were evident from the up-regulation of *TNFSF15*, *PTPN6*, *DUSP1*, *DUSP5* and *DUSP10* (Table S4: *ESI*†). *TNFSF15* was shown to exert incredible tumour suppressing effects through a regulatory axis of AMPK-LITAF-TNFSF15 and as such, the up-regulation of this gene is supportive of maslinic acid's role as a chemopreventive agent.<sup>35</sup> *PSMB9*, *UBD* and *TAP1* were positively regulated after 24 hours and these genes are representative of the classical pathway where this pathway usually regulates short half-life antigenic peptides generated from the ubiquitin-proteasome degradation system and provides rapid antigen detection.<sup>36,37</sup>

In response to tumorigenesis, the expression profile has proposed that maslinic acid is able to sensitize the cells to an alternative cell suicidal mechanism not only through the regulation of genes such as *PHLDA1*, *RYBP* and *BMP2* but also *via* the expression of *STAT1*. *STAT1* is one of the chief transcription factors which facilitate cellular homeostasis<sup>38,39</sup> through its participation in various aspects of cellular signaling by activating pathways such as the *STAT/IRF-1* signaling pathway. Hence, the results have suggested that the cells have undergone the *STAT/JAK* pathway which subsequently diverged into several cell death pathways which include death receptor/ligand dependent, transcription-dependent and necrosome-mediated cell death mechanisms.

## Conclusions

As maslinic acid was aptly described as a potential cancer chemopreventive agent, this compound has targeted other pathways in a simultaneous and gradual manner. This study has scrutinized and verified the prominent effects of maslinic acid at each time-point where a minimum of 12 hours incubation period was required for the compound to exert its full beneficial effects of suppressing oncogenes. Maslinic acid also strengthened the cells' immune response as well as its viral defense mechanism by restoring its antigenic presentation through the MHC antigen processing and presentation pathway.

## Declarations

### Ethics approval and consent to participate

Not applicable.

### Consent for publication

Not applicable.

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## Authors' contributions

WML Project development, Data collection, Bioinformatics analysis, Manuscript writing/editing; MS Bioinformatics analysis, Manuscript editing; HHG Bioinformatics analysis, Manuscript editing; YML Project development, Data analysis and interpretation, Manuscript writing/editing. YML conceived the entire study. All authors read and approved the final manuscript.

## Conflicts of interest

The authors declare no conflict of interest.

## Acknowledgements

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